

**DEPARTMENT OF CHEMICAL AND BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 331 QUIZ #2

This is a closed-book quiz. Write your name and answer the quiz problems related to your experiment. Turn in this question sheet with your answer.

Name: _____

Temperature Control Experiment

Problem 1 (35%) On a single well-labeled temperature vs. time plot, sketch the response curves for proportional control for three proportional gain (K_p) settings: **(a, 10%)** one that is large, **(b, 10%)** a second that is very small, and **(c, 10%)** a third that is intermediate. **(d, 5%)** Show the set-point temperature on the plot.

Problem 2 (40%) **(a, 10%)** Explain how derivative control reduces oscillation and stabilizes feedback control for a process. **(b, 10%)** For what classification of automatic control processes are PID controls useful? **(c, 20%)** Suppose that testing proportional-only control schemes for a process leads to large offset from the desired set point. Would you recommend adding integral control and/or derivative control to reduce this offset? If so, which one (or both) and why?

Problem 3 (25%) Label the block diagram for the generalized feedback control scheme given in Figure 1 (there are a total of five numbered labels to identify).

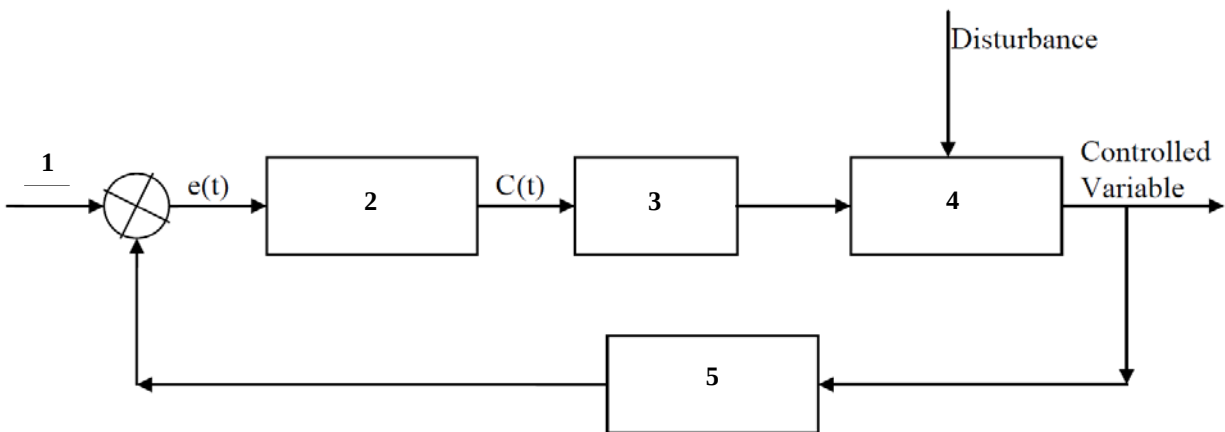


Figure 1. Feedback control system.